

Benjamin Cruse

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EDUCATION

Worcester Polytechnic Institute

BS in Robotics and Automation Engineering

MS in Robotics and Automation Engineering

Minor in Materials

Coursework: Motion Planning, Controls, Multi-Agent Systems, Dynamics, Safety Guarantees, Deep Learning

Worcester, MA 2022-26

May 2026

August 2026

September 2026

EMPLOYMENT

Research Apprenticeship: Automata Lab

(Blender, Isaac Sim, PyTorch, ROS 2)

May 2025 – Aug 2025

Contributed to research supporting Belief Space Control of Safety-Critical Systems Under State-Dependent Measurement Noise.

- Built a Blender-based virtual driving environment with procedurally generated tracks for training a lane-deviation model
- Created a configurable image generation and preprocessing pipeline for training PyTorch models
- Implemented a ROS 2 pipeline to process real-time simulated and physical camera video for model inference
- Created a digital twin in Isaac Sim to evaluate controllers on a simulated AWS DeepRacer

RESEARCH

Minimum Auction Spanning Task Assignment Hierarchy (MASTAH)

Master's Thesis

Aug 2025 – Sept 2026

- Developed MASTAH, a decentralized FJSP assignment algorithm for heterogeneous multi-agent task allocation
- Propagates task requests through an artificial hierarchy where each node maintains only a partial system view
- Searches local subtrees before expanding upward, using fitness auctions to select the closest best-suited viable machine
- Builds a KD-tree-like task hierarchy to limit combinatorial growth while keeping strong candidate machines nearby

Think Again: Augmenting Rigidly Safe Plans with General Safety Awareness Using LLMs

In preparation for submission to IEEE ICRA

Aug 2025 – Sept 2026

- Developed a hybrid LTL/LLM planner that preserves formal constraints while improving practical safety
- Converted LLM-generated hazard predictions into costs over weighted automata and navigation grids, prioritizing safer actions
- Implemented state-aware A* to balance path length, hazard exposure, and LTL task requirements
- Evaluated the system against baseline and LLM-based safety planners on simulated tasks and a Hello Robot Stretch 3

PROJECTS

Control Barrier Function Obstacle Avoidance on Ackermann Drive Robot

(ROS 2, CBFs, LiDAR, SLAM)

Mar 2026 – Apr 2026

- Demonstrated real-time obstacle avoidance on physical hardware using LiDAR-based perception and remote control computation
- Processed LiDAR scans to perform SLAM and detect nearby obstacles
- Applied Control Barrier Functions to enforce obstacle avoidance in an unknown environment
- Offloaded perception and control computation to a remote computer and streamed velocity commands to a physical AWS DeepRacer

TAMP Pick-and-Place Arm

(Genesis Sim, OMPL, PDDL)

Oct 2025 – Dec 2025

- Defined PDDL domain for structure building scenarios and created solver to plan and replan tasks on the fly
- Combined a sampling-based kinematic planner (BIT*) with gradient-based final pose correction
- Simulated 7 DOF Franka Emika Panda Arm to construct block structures

Forward and Inverse Kinematics & Inverse Dynamics Pipeline

(MATLAB, Numerical Optimization, RNE Algorithm)

Jan 2025 – May 2025

- Created platform agnostic FK/IK pipeline with singularity-avoidance from ground up utilizing screw theory
- Generated inverse dynamics via Recursive Newton-Euler algorithm to compute joint torques for desired position, velocity and acceleration
- Validated on simulated UR5 with 2 kg tip load along a quintic trajectory generated between 10 desired configurations. Computed torques and joint positions for 3,000 interpolated poses in under 0.75 seconds

Quadrotor Intercept Controller

(MATLAB, Optimal Control, Simulation)

Aug 2024 – Dec 2024

- Designed an LQR controller for a quadrotor to intercept a UAV with unknown trajectory using linearized 12-state dynamics.
- Enforced actuator limits of 3 N per rotor and modeled torque effects for realistic control authority.
- Achieved consistent intercepts of targets within 0.1 m at speeds of up to 4.8 m/s inside a 10x10x10 m flight volume.

TECHNICAL SKILLS

Robotics/Planning/Controls: Robot Control, MPC, LQR, PID, CLFs/CBFs, Dynamics, FK/IK, RNEA, SLAM, MCL, Sensor Fusion, EKF/UKF, MDPs/POMDPs, LTL/STL, Automata Theory, Task Planning, Motion Planning, Trajectory Optimization, PDDL, OMPL, Nav2

Perception/ML: Camera Calibration, Computer Vision, Data Processing, Dataset Curation, Automatic Data Labeling, OpenCV, PyTorch, CNNs, LSTMs, Transformers, Seq2Seq Models, Model Training & Inference

Modeling/Simulation: Blender, Onshape, Fusion 360, Isaac Sim, Genesis, RViz 2, Gazebo, URDF/SDF, KiCad

Languages/Tools: MATLAB, Python, C++, Java, Rust, Linux, Git, ROS & ROS 2, CMake, Docker, Docker Compose, Claude Code, Codex